

Enterprise VLAN & Trunking Lab

Phase 2 Report – Basic Switch Configuration

Project Information

Project Name: Enterprise VLAN & Trunking Lab

Phase: Phase 2 – Basic Switch Configuration

Simulation Software: Cisco Packet Tracer

Devices Configured: SW1, SW2

Status: Completed

1. Introduction

After completing the physical topology design in Phase 1, the next step was to prepare both Cisco switches for production use by applying a standardized baseline configuration. Establishing this baseline is a critical task during the deployment of enterprise network devices, as it ensures consistency, improves security, and provides a stable foundation for future network services.

During this phase, each switch was assigned a unique hostname, unnecessary DNS lookups were disabled, privileged access was secured using an encrypted password, console and virtual terminal (VTY) access were protected with authentication, and a legal Message of the Day (MOTD) banner was configured. The running configuration was then saved to startup memory to ensure persistence after a reboot.

2. Objectives

The objectives of this phase were to:

- Configure a unique hostname for each switch.
- Disable DNS lookup to improve CLI usability.
- Secure privileged EXEC mode with an encrypted enable secret.
- Encrypt locally configured passwords.
- Secure console access using password authentication.
- Secure VTY lines for future remote management.
- Configure a Message of the Day (MOTD) banner.
- Save the running configuration to non-volatile memory.

- Verify that all configurations were successfully applied.
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3. Devices Configured

Device	Hostname
Cisco 2960 Switch	SW1
Cisco 2960 Switch	SW2

Both switches received identical baseline configurations except for their unique hostnames.

4. Configurations Applied

4.1 Hostname Configuration

Each switch was assigned a descriptive hostname to uniquely identify the device within the network.

Device	Hostname
Switch 1	SW1
Switch 2	SW2

Meaningful hostnames improve administration, monitoring, troubleshooting, and documentation.

4.2 DNS Lookup Disabled

The following feature was disabled:

- IP Domain Lookup

Disabling DNS lookup prevents Cisco IOS from attempting to resolve mistyped commands as domain names, eliminating unnecessary delays while using the command-line interface.

4.3 Privileged EXEC Security

An encrypted **enable secret** password was configured on both switches.

Purpose:

- Protect administrative mode.
- Restrict configuration access.
- Store credentials securely using a cryptographic hash.

Unlike the legacy `enable password` command, `enable secret` provides significantly stronger protection and is the recommended method for securing privileged EXEC access.

4.4 Password Encryption

The `service password-encryption` feature was enabled.

Purpose:

- Prevent plaintext display of locally configured passwords.
- Improve confidentiality of console and VTY credentials.
- Reduce exposure during configuration reviews.

While this feature uses Cisco Type 7 encryption, which is not considered strong cryptography, it provides a basic layer of protection against casual observation and is commonly included in CCNA-level configurations.

4.5 Console Security

Console access was protected by:

- Configuring a console password.
- Enabling login authentication.
- Applying an inactivity timeout.

This ensures that anyone connecting physically to the switch must authenticate before gaining access to the Cisco IOS command-line interface.

4.6 Virtual Terminal (VTY) Security

The VTY lines (0–15) were configured with:

- Password authentication.
- Login enforcement.
- Idle session timeout.

These settings prepare the switches for secure remote management in future phases when SSH connectivity is introduced.

4.7 Message of the Day (MOTD)

A Message of the Day (MOTD) banner was configured on both switches.

Purpose:

- Notify users that the device is restricted to authorized personnel.
- Provide a legal warning against unauthorized access.
- Support organizational security policies and compliance requirements.

Such banners are commonly deployed in enterprise environments to reinforce acceptable use policies.

4.8 Configuration Persistence

The running configuration was successfully copied to startup configuration using:

- Running Configuration → Startup Configuration

This ensures that all settings remain intact after a device reload or power cycle.

5. Verification

The following verification activities were performed:

- Verified hostname configuration.
- Confirmed DNS lookup was disabled.
- Verified the presence of the encrypted enable secret.
- Confirmed password encryption was enabled.
- Verified console line configuration.
- Verified VTY line configuration.
- Confirmed the MOTD banner.
- Saved the running configuration.
- Reviewed the configuration using `show running-config`.

All verification steps completed successfully without configuration errors.

6. Skills Developed

This phase strengthened practical knowledge of:

- Cisco IOS command-line interface (CLI)
 - User EXEC mode
 - Privileged EXEC mode
 - Global Configuration mode
 - Line Configuration mode
 - Device identification
 - Local authentication
 - Password management
 - Configuration persistence
 - Enterprise switch initialization
 - Cisco configuration verification
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7. Best Practices Applied

The following industry best practices were followed during this phase:

- Assigned descriptive hostnames to all devices.
- Used `enable secret` instead of `enable password`.
- Disabled unnecessary DNS lookups.
- Protected console and remote access.
- Encrypted locally stored passwords.
- Configured an administrative warning banner.
- Saved configurations after implementation.
- Verified all configuration changes before proceeding.

These practices help improve the security, consistency, and maintainability of enterprise network devices.

8. Deliverables

The following deliverables were completed during Phase 2:

- Baseline configuration applied to SW1.
 - Baseline configuration applied to SW2.
 - Administrative access secured.
 - Configuration verified using Cisco IOS commands.
 - Startup configuration successfully saved.
 - Updated Packet Tracer project containing baseline configurations.
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9. Lessons Learned

Several key networking concepts were reinforced during this phase:

- Cisco IOS operates through multiple configuration modes, each serving a specific administrative purpose.
- Proper device naming greatly simplifies network management.
- Securing administrative access is one of the first priorities when deploying network infrastructure.
- Configuration changes remain temporary until explicitly saved.
- Verification commands are essential to confirm successful implementation and detect potential configuration errors.

These concepts form the foundation for all future switch configuration tasks.

10. Conclusion

Phase 2 successfully established a secure and standardized baseline configuration for both Cisco switches. Administrative access was protected, device identification was improved, and configuration persistence was ensured. Completing this phase provides a reliable foundation for implementing Layer 2 network segmentation in the next stage of the project.

With the baseline configuration complete, the switches are now ready for the creation of VLANs, assignment of access ports, and configuration of trunk links, enabling the logical separation of departmental traffic within the enterprise network.